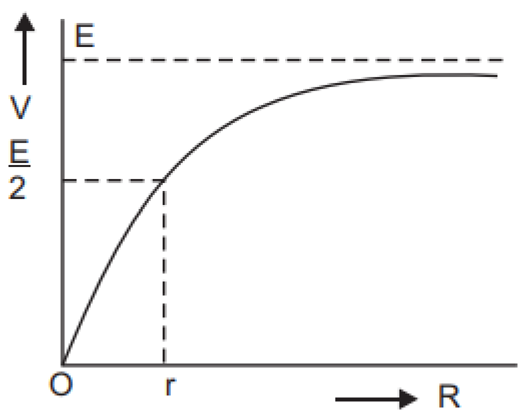
1^{st} PRE BOARD EXAM (2025-2026) LUCKNOW REGION

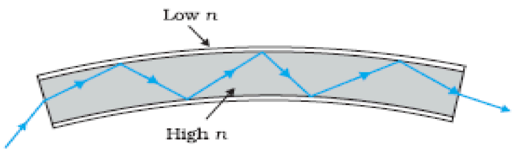
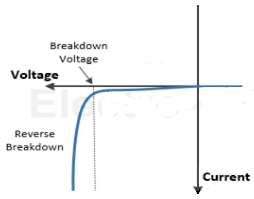
PHYSICS (Code No. 042) (SET1)

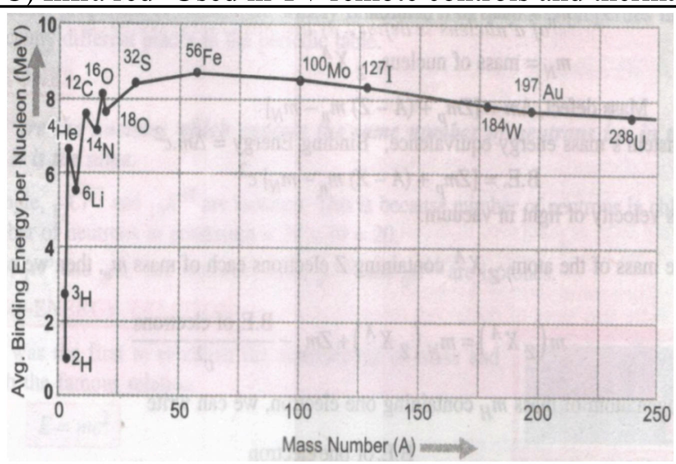
CLASS – XII

Time Allowed: 3 hours **Maximum Marks: 70**

MARKING SCHEME

	SECTION A	
Q.No	Question	Marks
1	(b) $Ea=2Ee$	1
2	(c) In A : V remains same and hence Q changes.	1
3	(d) 4B	1
4	(c) Equal to E	1
5	(a) $B = \mu H$	1
6	(b) Plate 1 will be positive and plate 2 negative	1
7	(d) $(i_o\, v_o \cos \varphi)/2$	1
8	(b) It is two terminal device which conducts current in one direction only.	1
9	(c) The beam of blue light would undergo total internal reflection.	1
10	(b) $\sqrt{(2h\nu_0/m)}$	1
11	(c) Most of the part of an atom is empty space	1
12	(b) VA > VB	1
13	(B)	1
14	(C)	1
15	(D)	1
16	(C)	1
	SECTION B	
17	(i) The emf E of a cell is independent of external resistance (R). (ii) (ii) The terminal p.d. $V=IR= ER/(r+R) = E/(1+r/R)$ On increasing R, V increases. When R = 0, $V\rightarrow 0$. When R = r, $V = E/2$ When $R \rightarrow \infty$, V = E.  	1+1

18	(i) (1,2) and (3,4) (ii) (1,3) and (2,4)]	1+1
19	 . Phenomenon : Total internal reflection <u>Working</u> : When a signal in the form of light enters at one end of the fibre at suitable angle, it undergoes repeated total internal reflections and finally comes out at the other end.	1+1
20(1)	**(i) Diverging lens.** **Reason :** $\frac{1}{f} = \left(\frac{\mu_g}{\mu_m} - 1\right)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$ As $\mu_m > \mu_g$ ⇨ $\frac{1}{f} = -ve$ ⇨ $f < 0$ **(ii) Converging lens.** **Reason :** $\frac{1}{f} = \left(\frac{\mu_g}{\mu_m} - 1\right)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$ As $\mu_m < \mu_g$ ⇨ $\frac{1}{f} = +ve$ ⇨ $f > 0$	2
20(II)	$f_{medium} = \frac{(a\mu_g - 1)}{\left(\frac{a\mu_g}{a\mu_w} - 1\right)} X\, f_{air} = \frac{(3/2-1)}{\left(\frac{3/2}{4/3} - 1\right)} X\ 22 = 4\ X\ 22 = 88\ cm$	2
21(1)	(i) Reverse bias **(ii)**  	1+1
21(II)	Depletion region : The small space charge region on either side of the p-n junction which becomes depleted from mobile charge carriers. is known as depletion region (i) Width of depletion region decreases in forward bias (ii) Width of depletion region increases in reverse bias	1+0.5+ 0.5
	SECTION C	
22	Mobility : Drift velocity per unit electric field is called mobility of electron i.e. $\mu = vd/$. It's unit is $m2\ /volt{-}sec$ $vd = -eE\ \tau/m$ $\mu = Vd/\ E = -e\ \tau/m$,	0.5+1.5 +1

	i.e. $\|\mu\|= e\,\tau/m$, At a constant potential difference, when the temperature is decreased the relaxation time increases as the mobility of electron is directly proportional to the relaxation time so the mobility also increases	
23	1. Given that $B_{axis} = \frac{1}{2\sqrt{2}} B_{centre}$ $\frac{\mu_0 I a^2}{2(a^2+r^2)^{3/2}} = \frac{1}{2\sqrt{2}} \frac{\mu_0 I}{2a}$ Solving for r, r = a = 1m.	3
24	Flux linked with coil S_1 , $\phi_1 = N_1 B_2$ x A, where A is effective/smaller area. $\phi_1 = N_1 \mu_0\ n_2 I A = N_1 \mu_0\ N_2 I A/l$ Also, $\phi_1 = M_1 I$ So, $M_1 = \mu_0 N_1 N_2 A/l$ Similarly , $M_2 = N_1 \mu_0\ N_2 A/l$	3
25	Refer to the example 12.3 of NCERT. $r=5.3*10^{-11}$m $v=2.2*10^{6}$m/s	1.5+1.5
26	Consider an equilateral triangle of side a Potential energy U= $kq1q2/\ a + kq2q3/\ a + kq1q3/\ a$ U= $k\ q\ .2q/\ a + k\ 2\ .nq/\ a + k\ q\ .nq/\ a$ According to question U=0 So $k\ q\ .2q/\ a + k\ 2q\ .nq/\ a + k\ q\ .nq/\ a = 0$, After solving n = −2/ 3	3
27(I)	Infrared radiations are produced by hot bodies and vibrations of molecules. They are referred to as heat waves because they are rapidly absorbed by water molecules and increase their thermal energy and heat them. Uses: Any two uses	1+1+1
27(II)	a) Gamma Rays : Used in cancer radiotherapy to destroy malignant cells.	1
	B) Microwaves. Used in radar systems and microwave ovens for cooking.	1
	C) Infra red -Used in TV remote controls and thermal imaging.	1
28		1+1+1

	Release of energy in fission & fusion : (i) When a heavy nucleus undergoes nuclear fission, the BE per nucleon of product nuclei is more than that of the original nucleus. This means that the nucleons get more tightly bound. Hence, there is release of energy. (ii) When two very light nuclei ($A \leq 10$) undergoes nuclear fusion, the BE per nucleon of product nucleus becomes more than that of the original lighter nuclei. This means that the nucleons in the final nucleus get more tightly bound. Hence, there is release of energy.	
	SECTION D	
29	(i) (d) spherical	1
	(ii) © called a wavefront	1
	(iii) (d) becomes broader	1
	(iv) © spherical wavefront and plane wavefront	1
30	(i) .(c) potential is maximum	1
	(ii) .(c) Insulators	1
	(iii) (d) Rate of thermal Generation of eleeton-hole pair	1
	(iv) (b) A Vacancy created when an electon leaves covalent bond	1
	SECTION E	
31(I) (i)	AC generator : It is a device which converts mechanical energy in to electrical energy. Principle : It is based on the principle of electromagnetic induction, i,e, whenever there is change in magnetic flux linked with a coil, an emf is induced in the coil Working : When the armature coil is rotated in a uniform magnetic field, effective area of coil ($A \cos\theta$) changes continuously due to which magnetic flux linked with it changes. Hence an emf is induced in the circuit and a current flows through the coil At any instant the magnetic flux linked with the coil $\phi = B A \cos\theta = B A \cos\omega t$ ⇨ Induced emf in the coil $e = -N \, d\phi/dt = -N \, d/dt(BA \cos\omega t)$ ⇨ $e = -NBA (-\omega \sin\omega t) = NBA \omega \sin\omega t$ Obviously, when $\sin\omega t = 1$ $e = e_{max} = e_0 = NBA \omega$ ⇨ $e = e_0 \sin\omega t$	3

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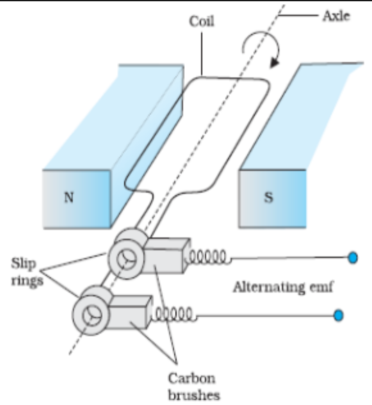
<tr><td></td><td>
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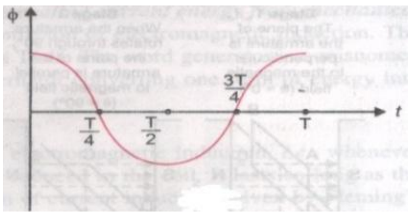
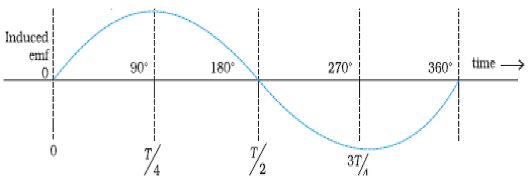
<tr><td>(ii)</td><td>(a)


(b)

</td><td>1+1</td></tr>
<tr><td>31(II)</td><td>Ans: i) X-resistor Y-inductor Z-capacitor
ii) Deduce expression for Z .
iii) Condition of resonance , $X_c=X_L$ so Z=R</td><td>1.5+2.5
+1</td></tr>
<tr><td>32(I)
(i)</td><td>Correct diagram</td><td>2</td></tr>
<tr><td>32(I)
(ii)</td><td>Advantages of reflecting telescope
(i) No chromatic aberration
(ii) No spherical aberration
(iii) Brighter image
(iv) large magnifying power
(v) High resolving power</td><td>2</td></tr>
<tr><td>32(I)
(iii)</td><td>m=(angle subtended at the eye by image)/(angle subtended at the eye by object)= f_o/f_e</td><td>1</td></tr>
<tr><td>32(II)
(i)</td><td></td><td>3</td></tr>
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Ans. Refraction through a glass prism :

Let a light ray is incident on the principal section ABC of a glass prism as shown

In quadrilateral AQNR,

□□ + 90^0 + □□□□□ + 90^0 = 360^0

□ □□ + □□□□□ = 180^0 ----------(1)

In triangle QNR,

□ + $□_2$ + □□□□□ = 180^0 ----------(2)

From (1) and (2)

□ + $□_2$ = □ --------(3)

Now, total deviation

□ = □□ − □□ + □□ − $□_2$□ = □□ + □□ − □□ + $□_2$□

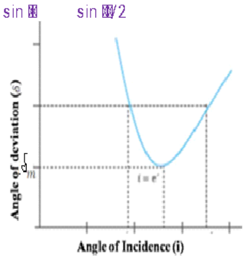
□ □ = □ + □ − □ ---------(4)

But when □ = $□_m$, □ = □ hence □ = $□_2$

□ from (3), 2□ = □ □ □ = □/2

From (4), $□_m$ = 2i − □ □ □ = □□ + $□_m$□/2

□ □ = $\frac{\sin □}{\sin □} = \frac{\sin\frac{□+□_m}{2}}{\sin □/2}$



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<tr><td>(ii)</td><td>

Given : $\mu_g = \sqrt{3}$, $A = 60^0$, i = ?

If the ray moves parallel to the base line, it means that,

$r_1 = r_2 = r$

As $r_1 + r_2 = A \Rightarrow 2r = 60^0 \Rightarrow r = 30^0$

$\mu_g = \frac{\sin sin\, i}{\sin sin\, r}$

$\Rightarrow \sqrt{3} = \frac{\sin sin\, i}{\sin sin\, 30^0}$

$\Rightarrow \sin sin\, i = \sqrt{3} \times \sin sin\, 30^0$

$= \sqrt{3} \times 1/2$

$= \sqrt{3}/2$

$\Rightarrow i = 60^0$

</td><td>2</td></tr>
<tr><td>33(I)</td><td>

(a). Gauss's Law : " Electric flux passing through any closed surface is $\frac{1}{\epsilon_0}$ times the total charge enclosed by that surface".

i,e, $\Phi_E = \oint \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$

</td><td>3+2</td></tr>
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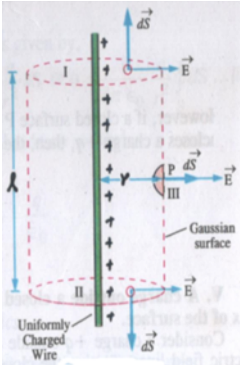


. Charge enclosed by Gaussian surface, [illegible]

At the part I and II of Gaussian surface [illegible] and [illegible] are [illegible], so flux through surfaces I and II is zero.

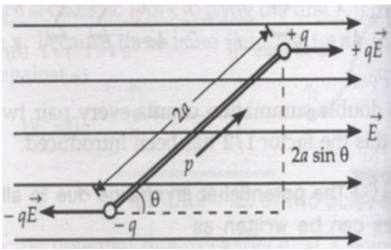
By Gauss's law, [illegible]

[illegible]

(b) Using $E = \frac{\lambda}{2\pi\varepsilon_0 r}$

Substituting the values we get $\lambda = 10^{-7}$ N/C

33(II) (i) Let an electric dipole of dipole moment (**p**) is placed in a uniform electric field (**E**) as shown in figure. 3+2



Force : Force on +q, $F_1 = qE$

Force on -q, $F_2 = -qE$

Hence net force on the dipole

$F = qE - qE = 0$

Torque :Two equal and opposite forces - qE and +qE forms a couple which tries to rotate the dipole.

Torque due to this couple

τ = either force X $\perp$ distance

$= qE \times 2a \sin\theta$

$= qE \times 2a \sin\theta$

$\Rightarrow \quad \tau = pE \sin\theta = (\mathbf{p}) \times (\mathbf{E})$

(ii) dipole moment of dipole,

	$P = qd = 6 \times 10^{-3}$ C × (10cm) = 6 × 10-3 C × (10/100) m = 6 × 10-3 C.m and angle between dipole , P and external electric field , E is , = 30° we know, torque = P.E sinθ or , $6\sqrt{3} = 6 \times 10^{-4} \times E \times \sin 30°$ or, $\sqrt{3} \times 10^{4} = E \times (1/2)$ or, $E = 2\sqrt{3} \times 10^{4}$ N/C	